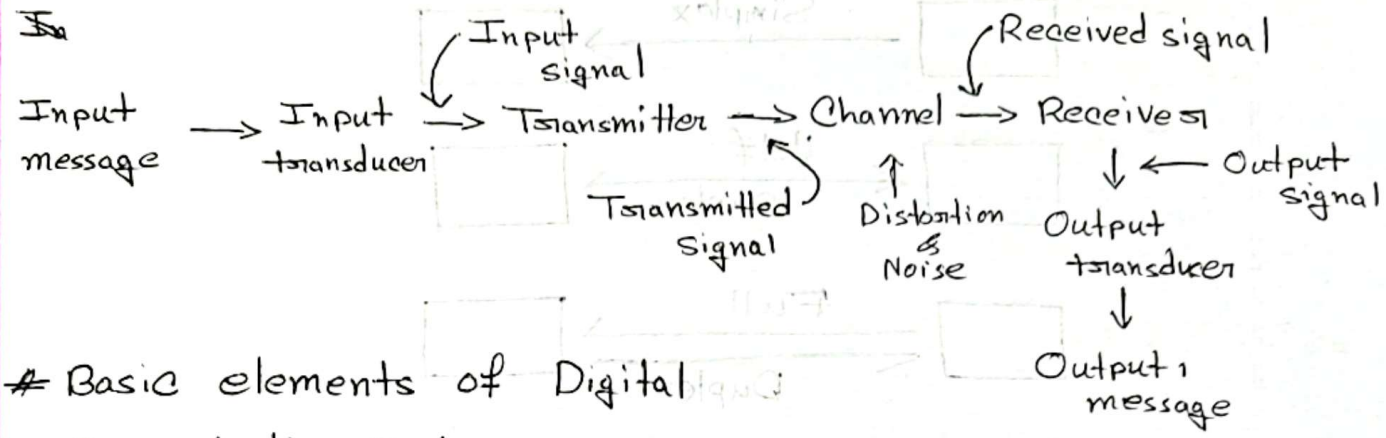
Digital Communication System



Basic elements of Digital Communication System–

Source → Any input signal (message)

Input transducer → Physical to electrical

Encoder → Error detection, ADC.

Modulator → Make suitable for transmission

Medium → Channel, kind of transmission media.

Demodulator → Retrive data from signal.

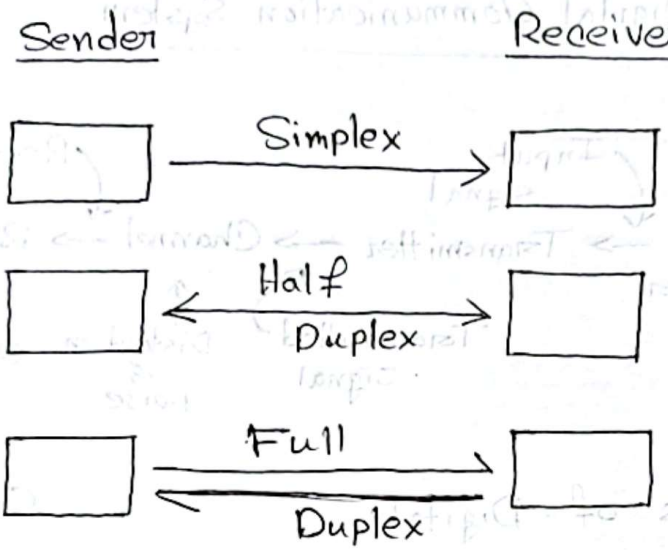
Decoder → Remove noise and reconstruct message.

Output transducer → electrical sound to signal, DAC.

Output → Destination of end user.

Based on Data-Flow (Communication Mode) –

1. Simplex – One way communication only.
2. Hdf-duplex – Two way communication but not at the same time.
3. Full-duplex – Two way communication at the same time.



Communication Channel →

1. Guided (Wired)
2. Unguided (Wireless)

Guided →

→ Twisted pair cable :-

Two individually insulated copper conductors twisted together.

Short distance, Low cost.

→ Shielded

→ Unshielded

LAN, Telephone, Internet Line.

→ Co-axial cable :-

A central copper conductor surrouded by a dielectric insulating layer, a metalic shielding layer and an outer jacket

TV, CCTV

—> Optical Fiber :-

Data transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection.

High Bandwidth, Low lass, Long distance.

Internet, Medical imaging, Submarine Communication.

Unguided ->

—> Radio wave:

Doesn't requires LOS (Line of sight).

Frequency -> 3 KHz - 300 GHz

TV Broadcasting, Navigation.

—> Microwave:

Requires LOS.

Frequency -> 1 GHz - 300 GHz

Satellite, Cellular communication

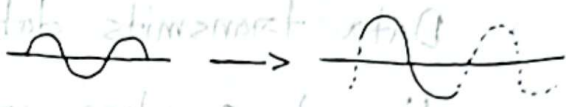
—> Infrared:

Short range wireless communication.

Frequency -> 300 GHz - 400 THz.

TV remote, Sensor.

Impairment Causes –

1. Attenuation 
2. Distortion 
3. Noise 

Noisy Channel Capacity / Shannon capacity –

$$C = B \log_2 (1 + SNR)$$

C = Channel Capacity (bps)

B = Channel Bandwidth (Hz)

SNR = Signal-to-Noise Ratio

if SNR is given in decibel (dB),

$$SNR = 10^{\frac{SNR_{dB}}{10}}$$

Bitrate, $= 2B \log_2 L$

B = Channel Bandwidth (Hz)

L = No. of signal level

Bitrate in bps

Prob 1

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. For this channel calculate the capacity C ?

=>

Given,

$$SNR = 0$$

Channel Capacity, $C = B\log_2(1+SNR)$

$= B\log_2(1+0)$

$= 0$ bps

(Ans)

Prob 2

SNR given in decible 15 dB and the channel bandwidth is 2MHz. Find the Shannon capacity.

=>

~~Shanno~~

$$SNR = 10^{\frac{SNR_{dB}}{10}} = 10^{\frac{15}{10}} = 31.623$$

$\therefore$ Shannon Capacity, $C = 2\times10^6 \times \log_2(1+31.623)$

$= 10055635.105$ bps

Ans

Prob 3

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What is the maximum bit rate?

=>

$$\text{Bitrate} = 2B \log_2 L$$

$$= 2 \times 3000 \times \log_2(2)$$

$$= 6000 \text{ bps}$$

$$= 6 \text{ kbps}$$

Ans

Prob 4

We need to send 265 kbps over a naseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

=>

As, $\text{Bitrate} = 2B \log_2 L$

$$\Rightarrow \log_2 L = \frac{\text{Bitrate}}{2B}$$

$$\Rightarrow \log_2 L = \frac{265 \times 1000}{2 \times 20 \times 1000}$$

$$\Rightarrow \log_2 L = 6.625$$

$$\therefore L = 2^{6.625} = 98.7 \text{ levels}$$

Ex – 3.35

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with 4 signal levels. The maximum bit rate?

=>

$$\text{Bitrate} = 2 \times 3000 \times \log_2 4 = 12000 \text{ bps}$$

$$= 12 \text{ kbps}$$

Ex – 3.38

A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. Whats the channel capacity?

=>

$$\text{Channel capacity, } C = B \log_2(1+SNR) = 3000 \log_2(1+3162)$$

$$= 34860 \text{ bps.}$$

Ex – 3.39

The $SNR_{dB} = 36$ and channel bandwidth 2 MHz. The Channel capacity?

=>

$$SNR = 10^{\frac{36}{10}} = 3981.07$$

$$\therefore \text{Channel capacity, } C = 2 \times 10^6 \times \log_2(1+3981.07)$$

$$= 23.91 \text{ Mbps.}$$

If the SNR is very high and assumed as

$SNR + 1 = SNR$,

For 2 MHz and $SNR_{dB} = 36$, Channel capacity?

=>

$$C = 2 \times 10^6 \times \frac{36}{3} = 24 \text{ Mbps} \quad \left[C = B \times \frac{SNR_{dB}}{3} \right.$$

Ex-3.41

We have a channel with 1 MHz bandwidth. The SNR for this channel is 63. What are the approximate bit-rate and signal level?

=>

Given,

$$C = 1 \times 10^6 \times \log_2(1+63)$$

$$= 6 \times 10^6 \text{ bps}$$

$$6 \times 10^6 = 2 \times 10^6 \log_2(L)$$

$$\Rightarrow \log_2(L) = \frac{6 \times 10^6}{2 \times 10^6}$$

$$\Rightarrow \log_2(L) = 3$$

$$\therefore L = 2^3 = 8 \quad \text{Ans}$$

Bitrate = 6×10^6 bps.

Level = 8.

Ans

Amplitude Modulation

Amplitude modulation -

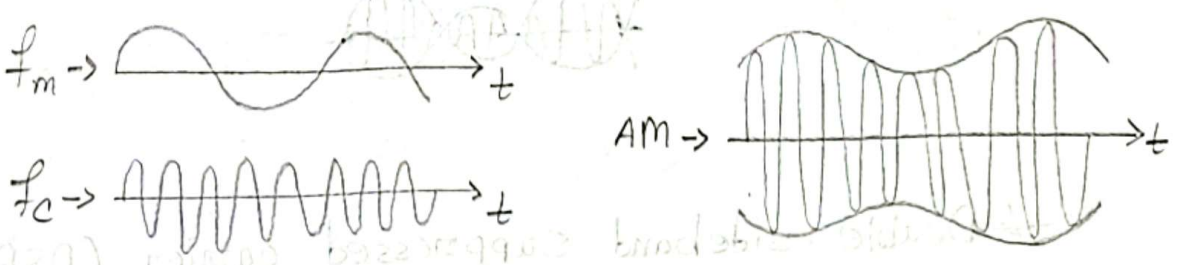
Amplitude modulation is a modulation technique where the amplitude of carrier signal will change according the amplitude of message signal but the frequency and phase will remain same in the carrier signal.

Message signal -> Low-frequency

Carrier signal -> High-frequency

$f_c >> f_m$

-> Low-frequency signals can't travel long distance, thats why Amplitude modulation is used.



Message signal, $m(t) = A_m \cos(2\pi f_m t) = A_m \cos \omega_m t$

Carrier signal, $c(t) = A_c \cos(2\pi f_c t) = A_c \cos \omega_c t$

∴ Amplitude modulation,

$$s(t) = [A_c + m(t)] \cos(2\pi f_c t) = [A_c + m(t)] \cos(\omega_c t)$$

$$\Rightarrow s(t) = [A_c + A_m \cos(\omega_m t)] \cos(\omega_c t)$$

$$\Rightarrow s(t) = A_c \left[1 + \frac{A_m}{A_c} \cos(\omega_m t)\right] \cos(\omega_c t)$$

$$\Rightarrow s(t) = [1 + \mu \cos(\omega_m t)] A_c \cos(\omega_c t)$$

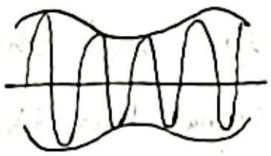
$$\Rightarrow s(t) = A_c \cos(\omega_c t) + A_c \mu \cos(\omega_m t) \cos(\omega_c t)$$

$$\therefore s(t) = A_c \cos(\omega_c t) + \frac{A_c \mu}{2} \cos(\omega_m + \omega_c)t + \frac{A_c \mu}{2} \cos(\omega_m - \omega_c)t$$

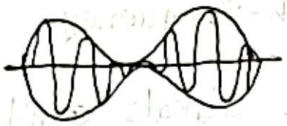
$$\mu = \frac{A_m}{A_c}$$

Modulation index, $\mu = \frac{Am}{Ac} = \frac{max - min}{max + min}$

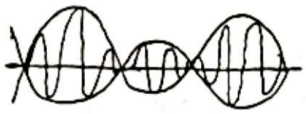
If, $\mu < 1$ => Under modulation



$\mu = 1$ => Perfect modulation



$\mu > 1$ => Over modulation



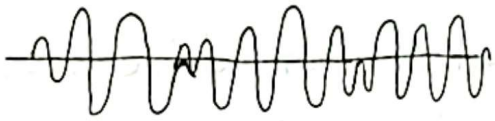
Double sideband suppressed carrier (DSBSC) ->

DSBSC modulation is a modulation technique where the carrier amplitude is varied according to the message signal but the carrier component is suppressed and only the two sidebands are transmitted.

$m(t) = A_m \cos(\omega_m t)$ → [Product Modulator] → $y(t) = m(t)\,c(t)$

$c(t) = A_c \cos(\omega_c t)$

DSBSC ->



Demodulation -

Demodulation is the process where the original message signal is extracted from the modulated carrier signal at the receiver.

Message signal → Modulator → Carried Signal → Demodulator → Original message Signal.

Types of Amplitude Demodulation -

1. Envelope Detector -

Uses diode RC circuit, works when carrier transmitted and signal is under or perfectly modulated.

-> Simple, cost-effective but not efficient.

2. Square Law -

-> Uses low pass filter to extract message.

-> Usually used for experiments.

3. Coherent / Synchronous Demodulation -

-> Demodulated suppressed carrier signals.

-> High efficiency & Noise immunity.

-> Complex circuitry.

Envelope Detector Circuit -

—> Works when diode forward biased.

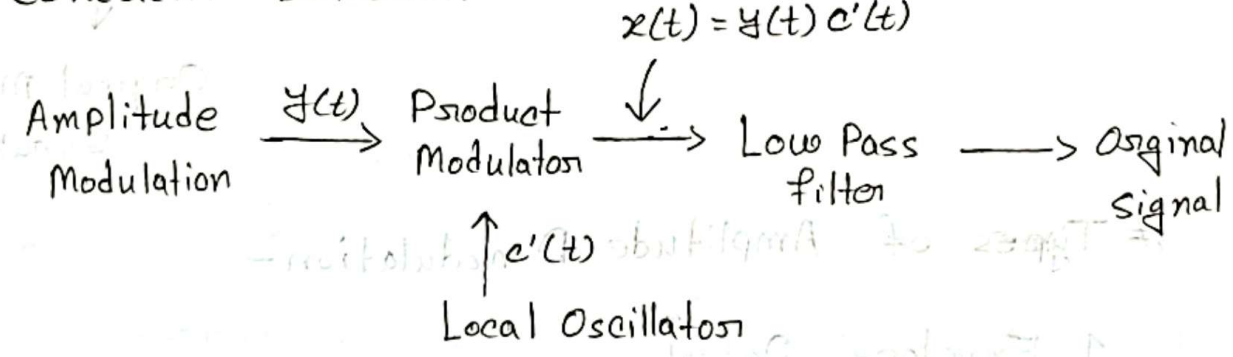
—> When voltage increases capacitor stores energy, then discharge when voltage drops.

—> Negative cycle is clipped.

—> Information loss.

—> $\frac{1}{f_c} << RC << \frac{1}{f_m}$, for optimum performance.

Coherent Detection -



Limitations of AM -

1. Large power waste in carrier.
2. Easily corrupted by noise.
3. Overmodulation is not suitable.
4. Limited bandwidth.
5. Bulky and expensive transmitter requirement.

Limitations of Amplitude demodulation –

1. Coherent detection is sensitive to phase synchronization and it has complex circuitry.
2. Envelope detector is noise affected easily, can't work with over modulated signal.

Importance of demodulation –

1. Communication between transmitter & receiver.
2. Recover original information.
3. Provide noise immunity.
4. Application in radio-tv broadcasting, satellite communication, internet data transmission, radar & navigation etc.

AM vs DSBSC –

Feature	AM	DSBSC
1. Carrier	Trasmitted	Suppressed
2. Detection	Envelop detector	Coherent detector
3. Complexity	Simple	Complex
4. Power efficiency	Low	High.

Angle Modulation

Angle modulation –

Angle modulation is a modulation process where the frequency or phase of the carrier signal changes with respect to message signal.

—> Frequency modulation:

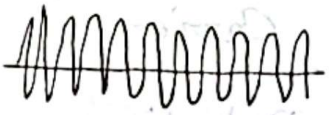
Frequency of carrier changes.

—> Phase modulation:

Phase of carrier changes.

Frequency modulation –

Frequency modulation is a modulation technique where the frequency of carrier signal will change according the frequency of message signal but the amplitude and phase will remain same in the carrier signal.

$f_m =$  $f_c =$ 

$y(t)$ 

FM Advantages –

1. Amplitude remains constant hence better noise immunity.
2. More efficient use of bandwidth.
3. Less power consumption than AM.
4. Prefered for radio broadcasting due to superior quality and capacity than AM.

FM Disadvantages–

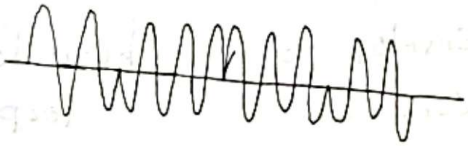
1. Works with higher bandwidth, which makes it costlier.
2. Antennas need to be closer for clear communication.
3. Complex transmitter-receiver settings.

Phase modulation –

Phase modulation is a modulation is a modulation technique where the phase of carrier signal will change according the phase of message signal but the amplitude and frequency will remain same in the carrier signal.

f_m = 

f_c = 

$f(t)$ = 

PM Advantages :

1. Fast operation
2. Higher efficiency than AM.
3. Circuit is less complicated compared to FM.

PM Disadvantages :

1. PM is less noise resistant than FM.
2. Poor signal to noise ratio.
3. Synchronization needed.
4. Not widely used.

Comparison -

Feature	AM	FM	PM
1. Modulated Parameter	Amplitude	Frequency	Phase
2. Noise immunity	Poor	Excellent	Good
3. Efficiency	Low	High	High
4. Complexity	Simple	High	High
5. Demodulation	Envelop or Coherent	Phase locked loop	Zero crossing detector

FM > PM > AM (Best sequence).

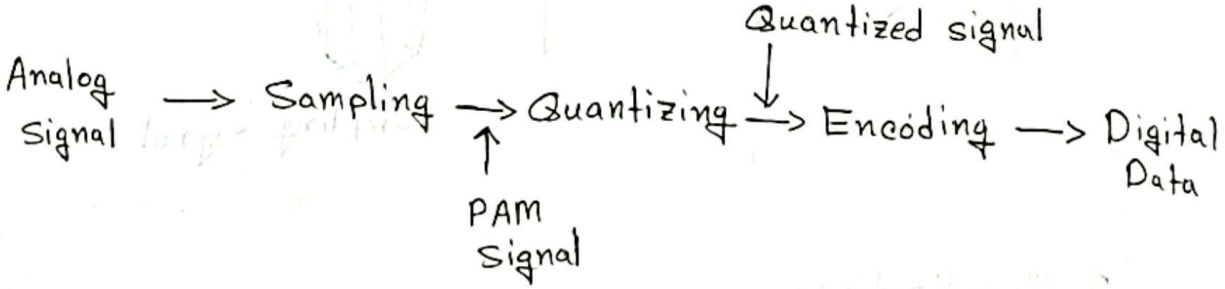
Pulse Code Modulation

Pulse code modulation or PCM converts the analog information into a binary sequence (1 and 0).

Input = 

Output = (PCM) 

Workflow ->

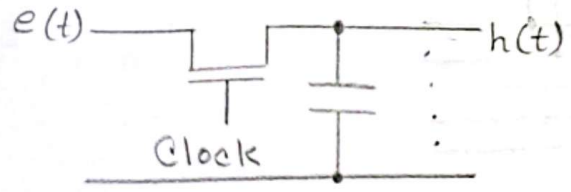


PCM has 3 main steps ->

1. Sampling
2. Quantizing
3. Encoding

1. Sampling :-

It is the process by which we convert continious time signal into discrete time signal by taking the signal values at some distinct points in time.

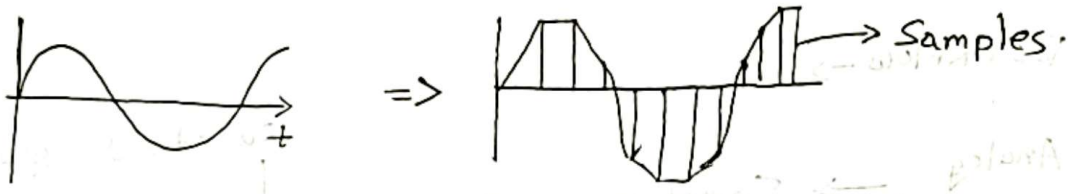


Sample and Hold circuit

Sampling rate = $\frac{1}{T} = f$ (Hz)

* Sampling Theorum :-

According to Nyquist criterion, the sampling rate must be twitch as the highest frequency of the analog signal. Otherwise, reconstructed signal will represent false image known as Aliasing.



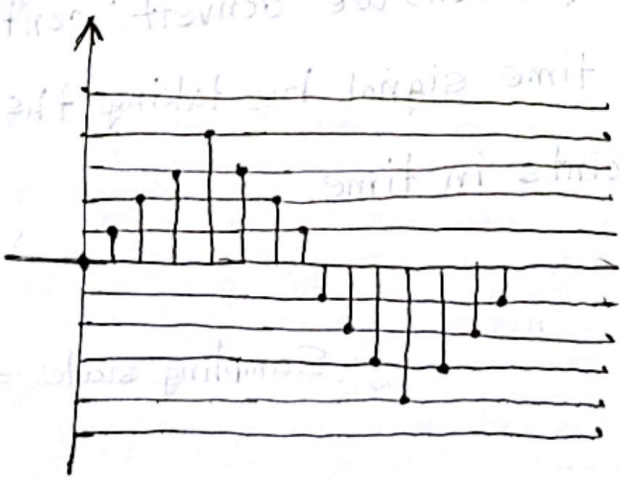
Sampling signal

2. Quantization :-

It is the process to represent a continuous valued signal with a limited set of discrete values.

* Quantization means converting sampled values into fixed discrete levels.

* More quantization levels result in better accuracy, Fewer quantization levels cause quantization errors.



Quantization of samples.

- Amplitude range of the signal is divided into L-discrete levels.
- Each sample is rounded off to the nearest level.
- More level = Less noise & distortion.
- 2^N quantization level for an N-bit quantizer.

3. Encoding :-

The process of converting each quantized level into binary code is called encoding.

If no. of quantization level = L

Then no. of bits per sample, $n = \log_2 L$

For example -

$L = 8$ & $n = 3$ bits

Then, levels encoded 000, 001, 010, ... 111 ->

(000, 001, 010, 011, 100, 101, 110, 111)

* Digital modulation techniques -

1. Amplitude Shift keying
2. Frequency Shift keying
3. Phase Shift keying
4. Binary phase Shift keying

ASK :-

Data is represented by changing the amplitude of the carrier signal.

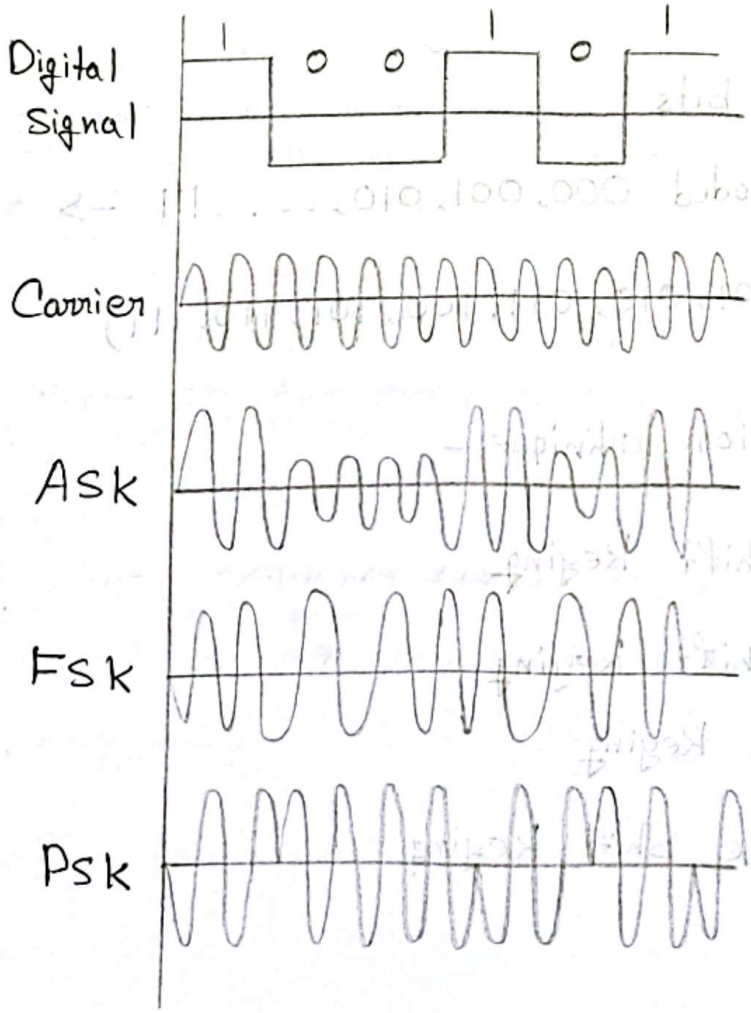
FSK :-

Data is represented by changing the frequency of the carrier signal.

PSK :-

Data is represented by changing the phase of the carrier signal.

—o—



Maths

Fosmulas –

Modulated signal, $= E_c (1 + m\cos\omega_s t)\cos\omega_c t$

$= E_c \cos\omega_c t + \frac{mE_c}{2}\cos(\omega_c + \omega_s)t + \frac{mE_c}{2}\cos(\omega_c - \omega_s)$

Here,

ω_c = Carrier component

ω_s = Message component

$\omega_s = 2\pi f_s$

$\omega_c = 2\pi f_c$

Power of carrier, $P_c = \frac{E_c^2}{2}$

Side band power, $P_s = \frac{1}{2} m^2 P_c$

Total power transmitted, $P = P_c + P_s$

Upper sideband frequency, $f_{USB} = f_c + f_s$

Lower Sideband frequency, $f_{LSB} = f_c - f_s$

Efficiency, $\mathcal{E} = \frac{P_s}{P} = \frac{m^2}{2+m^2}$

Power when resistor value is given, $P = \frac{(\text{rms value})^2}{R}$ $= \frac{(0.707 \times \text{Peak value})^2}{R}$

Modulation factor, $m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$

Maximum Amplitude of AM wave $= E_c + mE_c$

Minimum Amplitude of AM wave $= E_c - mE_c$

Sideband Amplitude $= \frac{mE_c}{2}$

Carrier Amplitude $= E_c$

Maximum or Minimum voltage, $V_{max/min} = \frac{\text{Peak-to-Peak}}{2}$

Ex-16.2

Problem-1

The maximum peak-to-peak voltage of an AM wave is 16 mV and the minimum peak-to-peak voltage is 4 mV. Calculate the modulation factor.

=>

Maximum voltage of AM wave is, $V_{max} = \frac{16}{2} = 8$ mV

Minimum voltage of AM wave is, $V_{min} = \frac{4}{2} = 2$ mV

$\therefore$ Modulation factor, $m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$

$= \frac{8-2}{8+2}$

$= 0.6$

Ex-16.5

Prob-2

An AM wave is represented by the expression:

$$V = 5(1 + 0.6\cos 6280t)\sin 211 \times 10^4 t \text{ volts.}$$

i) What are the minimum & maximum amplitudes of the AM wave?

ii) What frequency components are contained in the modulated wave and what is the amplitude of each component?

P.T.O

$\Rightarrow$

Equation =, $V = E_c(1 + m\cos\omega_s t)\sin\omega_c t$ - - - ①

Given,

Equation, $v = 5(1 + 0.6\cos\omega_s t)\sin\omega_c t$ - - - ②

Comparing ① & ② $\Rightarrow$

$E_c = 5$ V

$m = 0.6$

$f_s = \frac{\omega_s}{2\pi} = \frac{6280}{2\pi} = 1$ kHz

$f_c = \frac{\omega_c}{2\pi} = \frac{211 \times 10^4}{2\pi} = 336$ kHz

1) Minimum amplitude of AM wave, $= E_c - mE_c$

$= 5 - 0.6 \times 5$

$= 2$ V

Maximum amplitude of AM wave $= E_c + mE_c$

$= 5 + 0.6 \times 5$

$= 8$ V

11) Modulated wave contain,

Carrier frequency, $f_c = 336$ kHz

LSB frequency, $f_{LSB} = f_c - f_s = 336 - 1 = 335$ kHz

USB frequency, $f_{USB} = f_c + f_s = 336 + 1 = 337$ kHz

Carrier amplitude, $E_c = 5$ V

Sideband amplitude, $= \frac{mE_c}{2} = \frac{0.6 \times 5}{2} = 1.5$ V

Ex-16.8

Prob-3

A carrier wave of 500 watts is subjected to 100% amplitude modulation.

i) Power is sidebands.

ii) Power of modulated wave.

=>

i) Sideband power, $P_s = \frac{1}{2} m^2 P_c$

$= \frac{1}{2} \times 1^2 \times 500$

$= 250\ W$

ii) Power of modulated wave,

$P = P_c + P_s = 500 + 250 = 750\ W$

Ex-16.9

Prob-4

A 50 kW carrier is to be modulated to a level of

i) 80% ii) 10%

What is the total sideband power in each case?

=>

i) $P_s = \frac{1}{2} m^2 P_c = \frac{1}{2} (0.8)^2 \times 50 =$ ~~kW~~ 16 kW

ii) $P_s = \frac{1}{2} m^2 P_c = \frac{1}{2} (0.1)^2 \times 50 = 0.25\ kW$

Prob-5

An AM wave consists of the following components-

Carrier component = 5 V Peak value.

Lower sideband component = 2.5 V Peak value.

Upper sideband component = 2.5 V Peak value.

If the AM wave drives a 2 kΩ resistor, find the power delivered to the resistor by.

i) Carrier

ii) Lower sideband component

iii) Upper sideband component

iv) What is the total power delivered?

=>

i) $P_c = \frac{(0.707 \times 5)^2}{2000\ \Omega} = 6.25\ mW$

ii) $P_{LSB} = \frac{(0.707 \times 2.5)^2}{2000\ \Omega} = 1.562\ mW$

iii) $P_{USB} = \frac{(0.707 \times 2.5)^2}{2000\ \Omega} = 1.562\ mW$

iv) Total power, $P = P_c + P_{LSB} + P_{USB}$

$= (6.25 + 1.562 + 1.562)\ mW$

$= 9.374\ mW$

Ex-16.3

A carrier of 100 V and 1200 kHz is modulated by a 50 V, 1000 Hz sinewave signal. Find the modulation factor.

=>

$$\text{Modulation factor, } m = \frac{E_s}{E_c} = \frac{50\ V}{100\ V} = 0.5$$

Ex-16.6

A sinusoidal carrier voltage of frequency 1 MHz and amplitude 100 volts is amplitude modulated by sinusoidal voltage of frequency 5 kHz producing 50% modulation. Calculate the frequency and amplitude of lower and upper side band terms.

=>

$f_c = 1\ MHz = 1000\ kHz$

$f_s = 5\ kHz$ $\qquad E_c = 100\ V$

$m = 50\% = 0.5$

$f_{LSB} = f_c - f_s = 1000 - 5 = 995\ kHz$

$f_{USB} = f_c + f_s = 1000 + 5 = 1005\ kHz$

$$\text{Amplitude of sideband} = \frac{1}{2} m E_c$$

$$= \frac{1}{2} \times (0.5) \times 100$$

$$= 25\ V$$

Ex-16.7

A carrier wave of frequency 10 MHz and Peak value 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6V. Determine -

I) Modulation factor

II) Sideband frequencies

III) Amplitude of sideband components

IV) Draw the frequency spectrum

=>

$E_c = 10\ V$

$E_s = 6\ V$

$f_c = 10\ MHz$

$f_s = 5\ kHz = 0.005\ MHz$

I) $m = \frac{E_s}{E_c} = \frac{6}{10} = 0.6$

II) $f_{LSB} = f_c - f_s = 10 - 0.005$

$= 9.995\ MHz$

$f_{USB} = f_c + f_s = 10 + 0.005$

$= 10.005\ MHz$

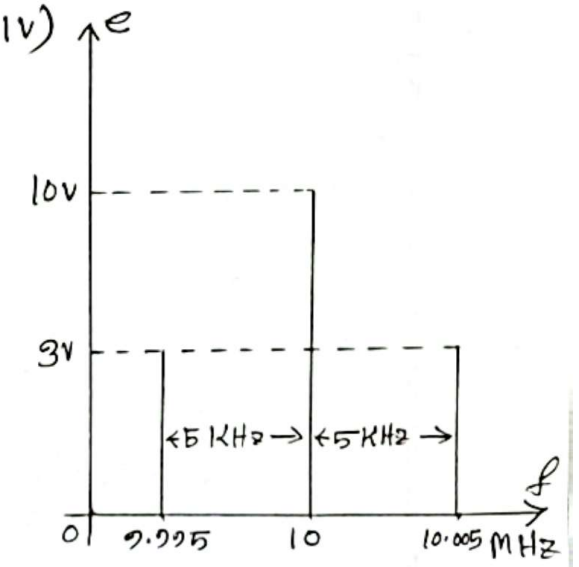
III) Amplitude of each sideband

$= \frac{mE_c}{2}$

$= \frac{0.6 \times 10}{2}$

$= 3\ V$

IV)



H.W

Q-1 # Problem 4 ->

A 50 kW carrier is to be modulated to a level of i) 80% ii) 10%. What is the total sideband power in each case?

=>

Given,

Carrier power, $P_c = 50$ kW

i) modulation, $m = 80\% = 0.8$

$\therefore$ Sideband power, $P_{SB} = \frac{1}{2} m^2 P_c$

$= \frac{1}{2} \times (0.8)^2 \times 50$

$= 16$ kW Ans

ii) modulation, $m = 10\% = 0.1$

$\therefore$ Sideband power, $P_{SB} = \frac{1}{2} m^2 P_c$

$= \frac{1}{2} \times (0.1)^2 \times 50$

$= 0.25$ kW Ans

Q-2

An AM wave is represented by the expression:

$V = (10 + 6\cos 4356t)\sin 133\times10^3 t$ Volts, Relate this with Standard AM wave equation and find values of each variable.

=>

Given,

$$V = (10 + 6\cos 4356t)\sin 133\times10^3 t$$

Standard equation,

$$V = E_c(1 + m\cos\omega_s t)\sin\omega_c t \quad ----①$$

$$\therefore V = 10\left(1 + \frac{6}{10}\cos 4356t\right)\sin 133\times10^3 t \quad ----⑪$$

∴, Comparing ① & ⑪ =>

$$E_c = 10 \text{ V}$$

$$m = \frac{6}{10} = 0.6$$

$$\omega_s = 4356 \text{ rad/s}$$

$$\omega_c = 133\times10^3 \text{ rad/s}$$